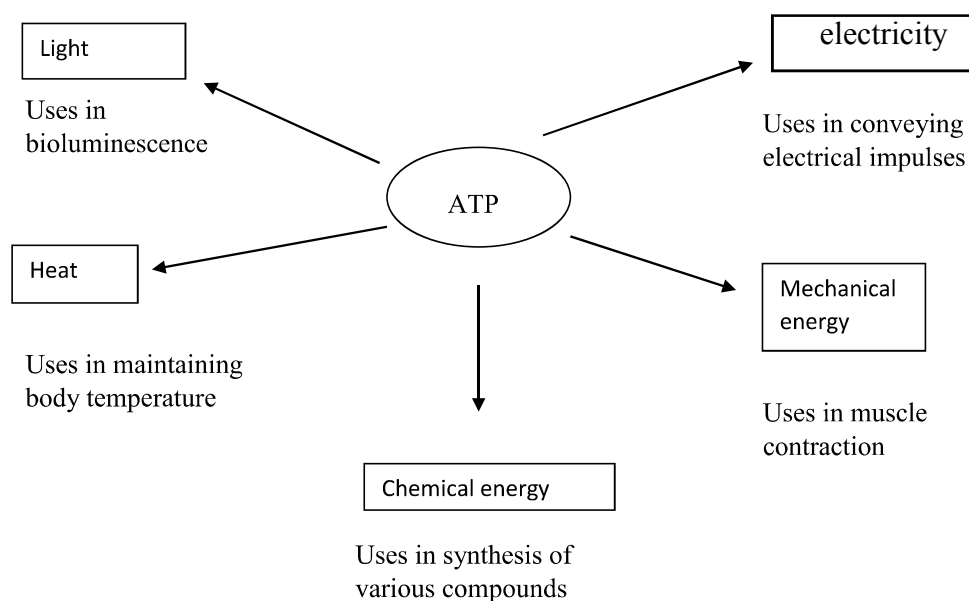




The role of Enzymes in regulating metabolic reactions

An enzyme is a macromolecule, which acts as a biological catalyst. Enzymes are produced in living cells/

General characteristics of an enzyme:

1. Most of the enzymes are globular proteins.
2. Enzymes are biological catalysts. They lower the activation energy of the reaction they catalyze (increases the rate of reaction).
3. Most enzymes are heat liable/ sensitive
4. Their presence does not alter the nature or properties of the end products of any reaction.
5. Enzymes are highly specific to the substrate (substrate specific)
6. Most enzyme catalyzed reactions are reversible.
7. The rate of enzyme activity is affected by pH, temperature and substrate concentrations.
8. They are not being used up during the reaction.
9. Enzymes possess active sites where the reaction takes place.
10. Some enzymes need non-proteinous components to catalyse the reaction which are known as cofactors.

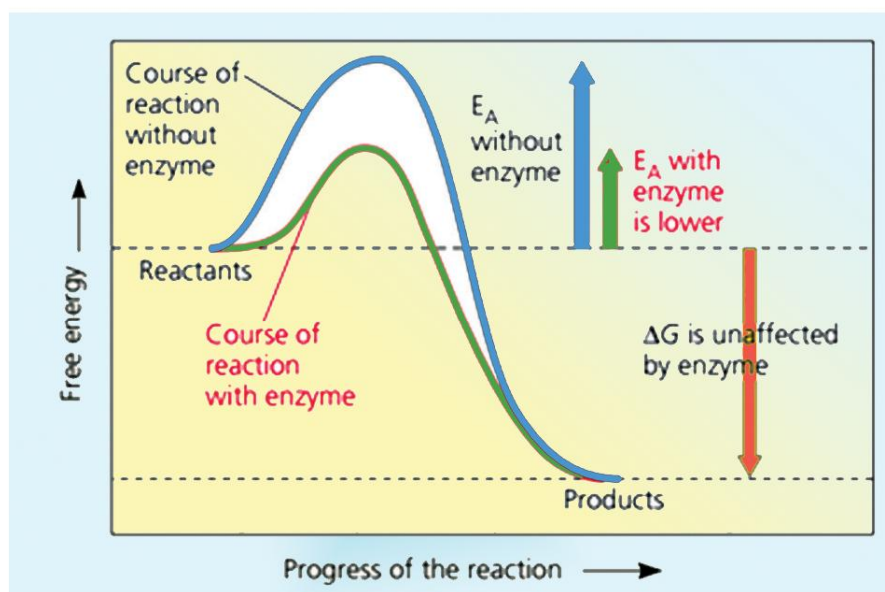


Fig 2.30 - The relationship between activation energy and the enzyme

The mechanisms of enzyme action

The reactant and enzyme acts on is referred to as the substrate. The enzyme binds to its substrate forming enzyme-substrate complex. While enzyme and substrate form their complex, catalytic action of the enzyme converts the substrate to the product.



The reaction catalyzed by each enzyme is very specific. The specificity of an enzyme results from its shape. The substrate binds to a specific region of the enzyme. This region is called the active site. The active site is formed by only a few amino acids. Other amino acids are needed to maintain the shape of the enzyme molecule. The shape of the active site is complementary to the shape of the specific substrate of the enzyme, and hence important in the substrate specificity of the enzyme. The shape of the active site of an enzyme is not always fully complementary to its substrate. As enzymes are not rigid structures, the interactions between substrate and active site may slightly change the shape of the active site, so that the substrate and the active site become complementary to each other. This is called induced fit mechanism. The tight fit not only brings the substrate molecules and the active site close to each other, but also ensures the correct orientation of the molecules to help the reaction to proceed and catalyzes the conversion of substrate to product. Thereafter, the product departs from the active site of the enzyme. The enzyme is then free to take another substrate molecule into its active site.

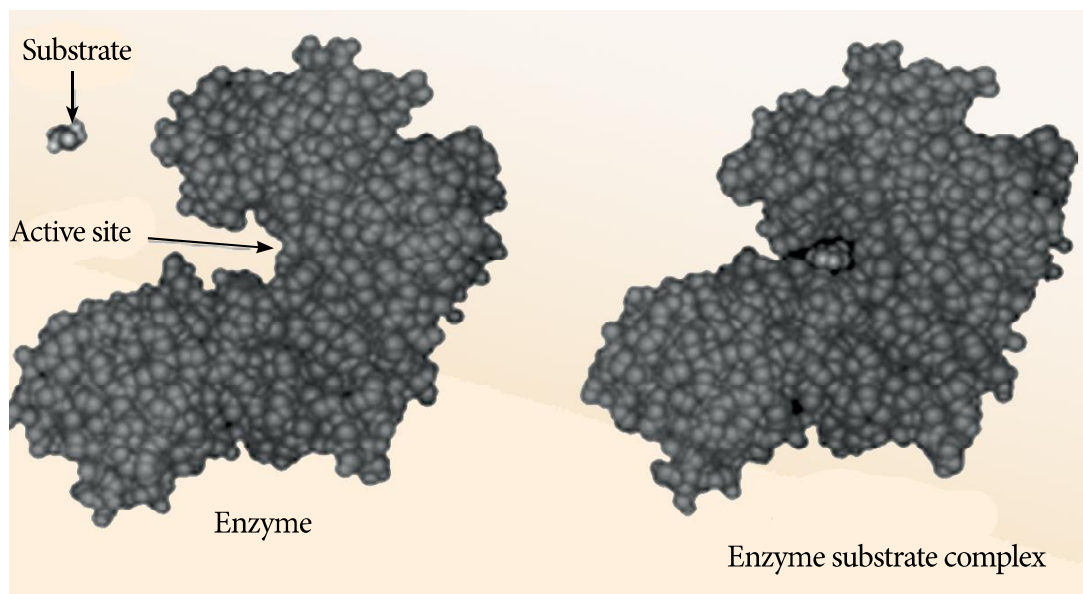


Fig 2.31: Induced fit between an enzyme and its substrate

Cofactors

Non-proteinous components which are essential for the catalytic activities of certain enzymes are called cofactors.

These cofactors bind to the enzymes in two ways. Some tightly bind and remain permanently and others loosely bind temporarily. Loosely bound cofactors are reversible under certain circumstances.

Organic cofactors are called co-enzymes. e.g. derivatives of vitamins e.g. NAD, FAD and biotin

Inorganic co-factors – e.g. Zn^{2+} , Fe^{2+} , Cu^{2+}

Factors affecting the rate of enzymatic reactions

1. Temperature
2. pH
3. Substrate concentration
4. Enzyme concentration
5. Inhibitors

Temperature

Increase in temperature increases molecular motion. Therefore the speed of the moving molecules of both enzymes as well as the substrate will be accelerated. This will enhance the colliding probability for both enzyme active sites and substrate molecules. More collision between the enzyme active sites and substrate molecules generate greater chances for the reaction to occur. This can continue up to a certain

point, after which there is a rapid decline in enzyme activity. This point is referred to as optimum temperature. This may vary from organism to organism.

e.g. most of the human enzymes have optimum temperature around the body temperature (35°C - 40°C). Optimum temperature of bacteria in hot springs is about 70°C .

When the temperature increases beyond the optimum temperature, the hydrogen bonds, ionic and other weak chemical bonds of enzyme active sites may be disrupted. This will result a change in the shape of the active site of enzyme which will alter the complementary nature of the active site of enzyme molecules. Therefore, the complementary binding of enzyme active sites and substrate molecules will be prevented. The above event is called as denaturation of enzyme molecules.

Therefore the rate of enzyme catalyzed reaction will start to decline when the temperature increases beyond the optimum temperature and stops completely at certain temperature, although rate of collision will keep on increasing.

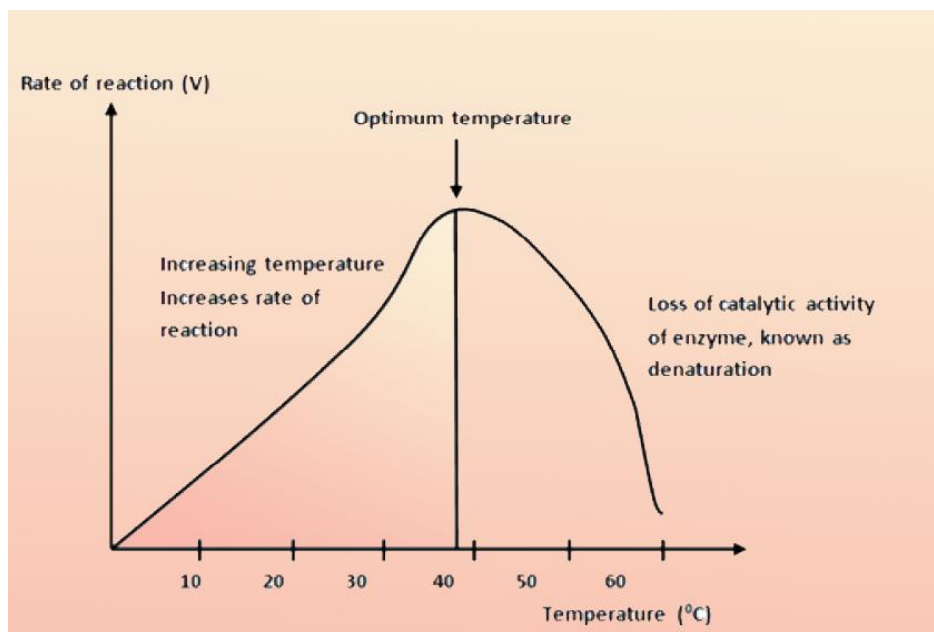


Fig -2.32 The graph of Rate of reaction (V) vs Temperature(T)

pH

Enzymes function most efficiently within a certain pH range despite maintaining temperature of the environment constant.

The narrow range of pH in which a particular enzyme catalyzed reaction takes place is named as the pH range. The pH at which the highest rate of reaction occurs is the optimum pH of the enzyme. The alteration in pH above or below the optimum pH may lead to decline in enzyme activity. This is due to the alteration of chemical bonds involving in formation of enzyme substrate complex. In most enzymes optimum pH range is 6-8, but there are exceptions. Pepsin works best at pH 2 and optimum pH for Trypsin is 8.

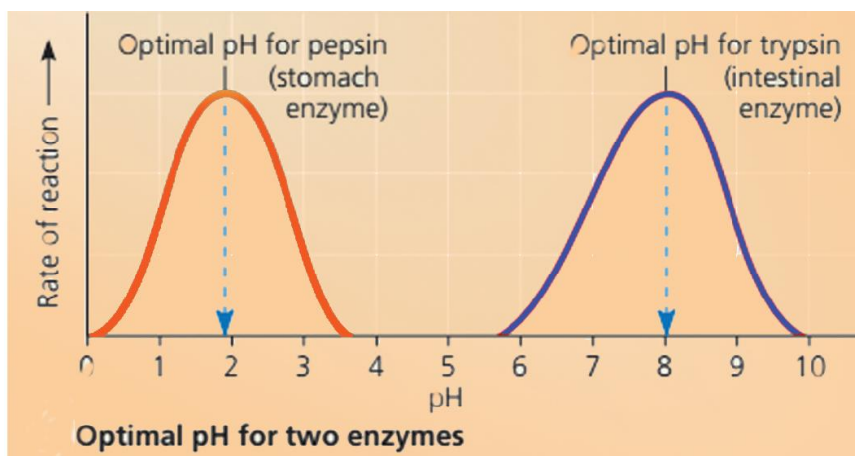


Fig 2.33- Rates of reaction of two enzymes at various pH values

Substrate concentration

Increasing substrate concentration increases the probability of collision between the enzyme and substrate molecules with correct orientation. However the enzyme molecules will be saturated after a particular concentration and therefore there will not be any further increase in the rate of reaction.

Enzyme inhibitors

Certain molecules or ions selectively bind permanently or temporarily to the enzyme molecules and prevent them from forming enzyme-substrate complex. These substances are called inhibitors.

They are either binding reversibly with weak interactions or binding irreversibly through covalent bonds.

e.g. Irreversible inhibitors: toxins, poisons

Reversible inhibitors- drugs used against microbes

Competitive inhibitors

Most of these are reversible inhibitors. These chemicals resemble the shape and nature of the substrate. Therefore they compete with the substrate selectively for the active site of certain enzymes. As a result of the above, the number of active sites available for the enzymes may decline and therefore reduces the rate of enzyme catalyzed reactions.

The above situation may be reversed by increasing the substrate concentration.

e.g. Protease inhibitor of drugs against HIV.-change

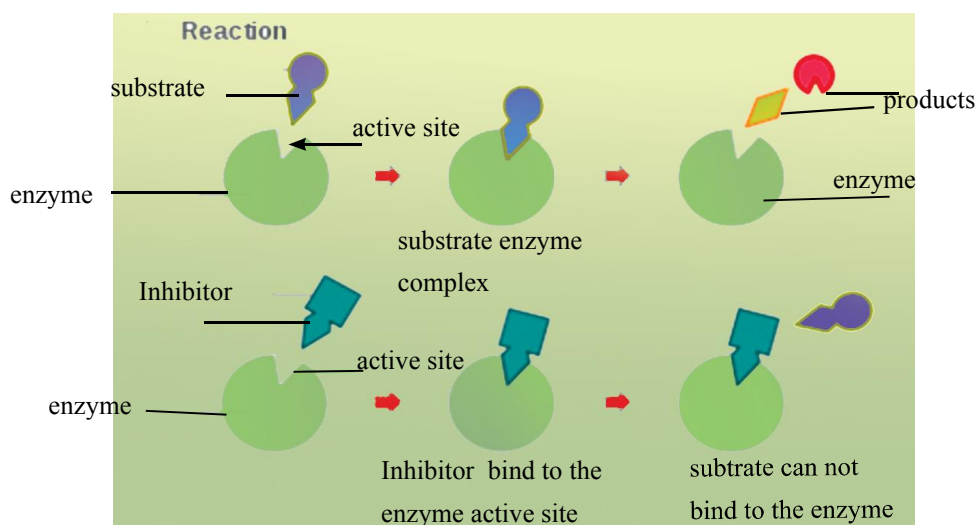


Fig 2.34: Competitive inhibitors

Non-competitive inhibitors

These chemicals do not compete with substrate molecules. They interrupt enzymatic reaction by binding to a part of the enzyme other than the active site. This causes the enzyme molecule to change its shape in such a way that the active site becomes less effective for the formation of enzyme substrate complex.

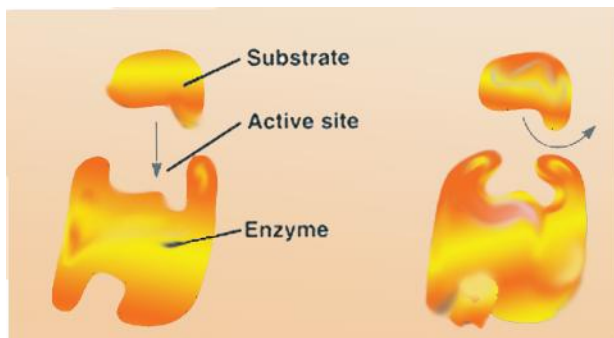


Fig 2.35: noncompetitive inhibitors

Regulation mechanism of enzymatic activity in cells

Allosteric regulation of enzymes

In many cases, the molecules that naturally regulate enzyme activity in a cell behave like reversible non-competitive inhibitors. Regulatory molecules (either activators or inhibitors) bind to specific regulatory sites elsewhere (other than the active site) of the molecule via non-covalent interactions and affect the shape and function of the enzyme. It may result in either inhibition or stimulation of an enzyme activity.

a.) Allosteric activation and inhibition

Most enzymes regulated by allosteric regulation are made from two or more subunits. Each sub unit composed of a polypeptide chain with its own active site. The entire complex oscillates between two different shapes one catalyzing active and other inactive. In this two forms regulatory molecules bind to a regulatory site called allosteric site, often located where subunits join.

When an activator binds with this regulatory site, stabilizes the shape with functional active sites. Whereas the inhibitor binds with the regulatory site, it stabilizes the inactive form of enzyme. Subunits of enzyme arranged in a way through which they transmit the signals quickly other subunits. Through the interaction of subunits even a single activator or inhibitor molecule that bind to one regulatory site will affect the active site of all sub units. e.g. ADP function as allosteric activator bind to the enzyme and stimulates the production of ATP by catabolism. If the supply of ATP exceed demand catabolism shows down as ATP bind to the same enzyme as inhibitor.

b.) cooperativity

This is another type of allosteric activation. Binding of one substrate molecule can stimulate binding or activity at other active site. Thereby increase the catalytic activity. e.g. hemoglobin (not an enzyme) is made up of four subunits each with an O₂ binding site. The binding of a one molecule of O₂ to one binding site increases the affinity for O₂ of the remaining binding site.

c.) Feedback inhibition

In feedback inhibition, a metabolic pathway is stopped by the inhibitory binding of its end product of a process to an enzyme. Thereby limit the production of more end products than required and thus wasting chemical resources.

Feedback inhibition

Feedback inhibition is an essential process regulates the end products produced in metabolism.

e.g. ADP function as allosteric activator and stimulates the production of ATP during the catabolism.

In case ATP supply exceeds demand, catabolism slows down as ATP molecules function as allosteric inhibitor.

Energy needed for all living processes is obtained directly from ATP. ATP is mainly produced by a process called cellular respiration, in living cells.

Photosynthesis as an energy fixing mechanism

Photosynthesis

Photosynthesis is a metabolic process by which light energy is trapped and converted to chemical energy. Chemical energy is stored in chemical bonds of carbohydrates, fats, oils, and proteins. All life on Earth depends on photosynthesis either directly or indirectly. Photosynthesis also occurs in algae and certain prokaryotes.

Global importance of photosynthesis;

- All life on earth depends on photosynthesis, directly or indirectly
- Fulfill both carbon and energy requirements of organisms
- Provide O₂ for respiration of aerobic organisms
- Maintain O₂ and CO₂ balance in the atmosphere
- Production of fossil fuel
- Maintenance of global temperature

During photosynthesis CO₂ is reduced by the H of H₂O and simple sugars are made using light energy. In eukaryotic photosynthetic cells, chloroplasts are the sites of photosynthesis.

Process of photosynthesis consists of two main stages and they are integrated.

- Light-dependent reaction
- Calvin cycle

There are two types of photosynthetic mechanisms (path ways) based on the number of C atoms of the first stable product of the CO₂ fixation.

- C₃ Mechanism – No of C atom of the first stable compound is three
- C₄ Mechanism – No of C atom of the first stable compound is four

Light-dependent reactions of photosynthesis take place in the membrane system of thylakoids. They are flattened fluid-filled sacs, which form stacks called grana at intervals. Chlorophylls, carotenoids and electron acceptors are located on this membrane system of thylakoids.

Stroma is a gel like structure containing soluble enzymes and other chemicals, which is the site of the Calvin cycle.

Photosynthetic pigments are substances which absorb visible light. In a leaf we see green colour because chlorophylls absorb violet, blue and red light and therefore, they transmit and reflect green colour. Different pigments absorb different wavelengths of light. In chloroplast, there are two types of chloroplast pigments such as chlorophylls and carotenoids. Chlorophyll a is the key light capturing pigment and they participate directly in the light reaction of photosynthesis.

According to the action spectrum, chlorophyll a is more effective for blue and red light. Chlorophyll b and carotenoids (carotenes and xanthophylls) are effective in absorption of specific range wavelengths of corresponding to different colours.

Other important function of some carotenoids is photoprotection. Photoprotection is absorption and dissipation of excessive light energy, if not that excessive light may cause damage to the chlorophylls or interact with oxygen and form reactive oxidative molecules which are dangerous to the cell.

Absorption spectrum

An absorption spectrum is a graph of the relative amounts of light absorbed at different wavelengths by a pigment.

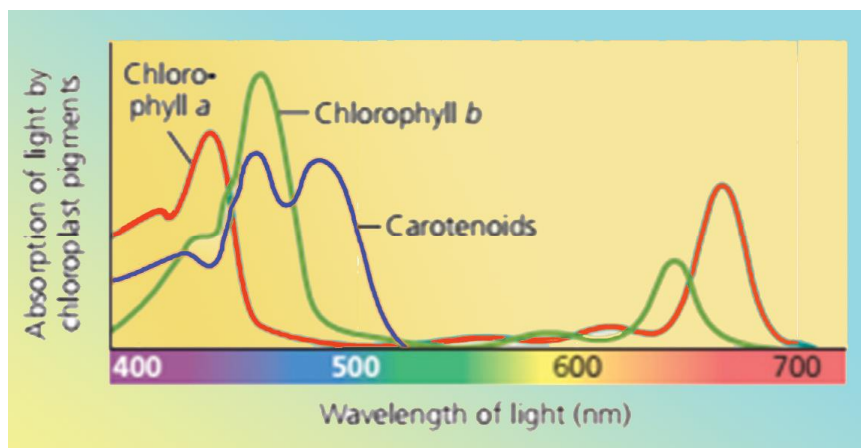


Fig. 2.36: Absorption spectrum

Action spectrum

An action spectrum is a graph showing the effectiveness of different wave lengths of light in stimulating the photosynthesis.

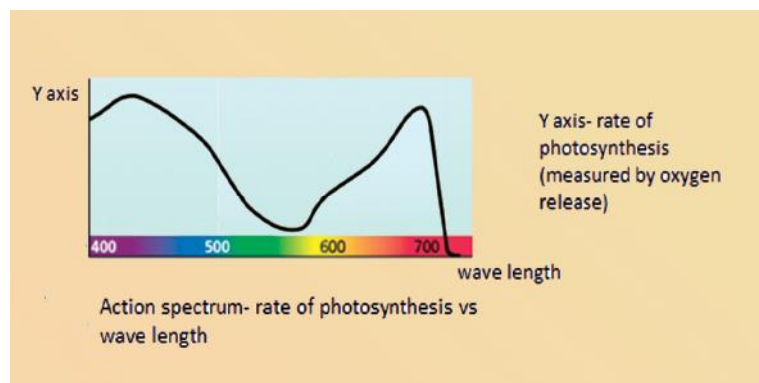


Fig 2.37: Action spectrum

Excitation of chlorophyll by light

When a molecule of chlorophyll or other photosynthetic pigment absorbs light it becomes excited. The energy from the light is used to boost electrons to a higher level and become positively charged. The excited state is unstable and returns to their original lower energy state. The excited electrons may pass through several electron carriers until they reach the final electron acceptor.



Therefore chlorophyll is oxidized and electron acceptor is reduced.

Photosystems

Chlorophyll molecules, other organic molecules and proteins are organized into complexes in the thylakoid membrane of chloroplasts. They are called photosystems.

A photosystem contains a reaction centre complex and light harvesting complexes. The reaction centre complex also contains a primary electron acceptor.

There are two types of photosystems found in the thylakoid membrane. They are Photosystem I (PS I) and photosystem II (PS II). In the PS I the chlorophyll a molecule is known as P700 since they absorb light at 700nm wave length effectively. In the PS II the reaction centre contains a chlorophyll a molecule which is known as P680 which absorbs light having a wavelength of 680 nm.

Light-dependent reaction /Light reaction photosynthesis

Linear electron flow

Light is absorbed by the photosynthetic pigments and synthesize ATP and NADPH due to the excitation of Photosystem I and Photosystem II which are embedded in the thylakoid membrane of chloroplast. The key to this energy transformation is a flow of the electron in one direction through the photosystems and other molecular components built in the thylakoid. This process is called linear electron flow.

The striking of photons of light on the pigments results in the excitation of electrons from the photosystem II to the higher energy state.

These electrons will be accepted by the primary electron acceptor of photosystem II. Splitting of water takes place as a result of an enzyme catalyzed reaction and yields O₂ (g), H⁺ ions and electrons.

Electrons released as a result of hydrolysis may neutralize excited photosystem II (P680).

Striking of photons of light on the pigments results in the excitation of electrons from photosystem I (P700) to the higher energy state. Excited electrons will be accepted by a primary electron acceptor of PSI.

Excited electrons of PS II at primary electron acceptor of PS II will pass through an electron transport chain to PS I and neutralize the excited PS I. The energy released due to the passage of electrons from higher energy state to lower energy result in the synthesis of ATP. This is known as photophosphorylation. Excited electrons of PS I at primary electron acceptor of PSI will pass through an electron transport chain and

reduce NADP and yield NADPH. The reduction of NADP is catalyzed by an enzyme called NADP reductase.

Cyclic electron flow

This occurs in photosystem I but not in Photosystem II. Here some photoexcited electrons use alternative cyclic pathway. This produces ATP but not NADPH and Oxygen is released.

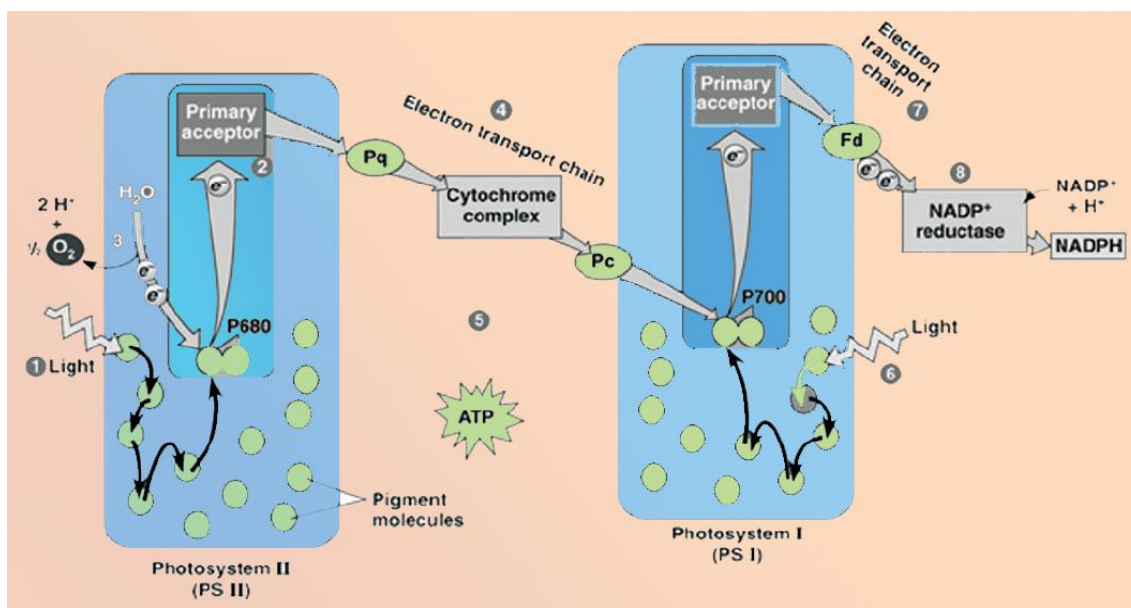


Fig 2.39 : Linear electron flow in the light reaction of photosynthesis

The Calvin cycle

The Calvin cycle takes place in the stroma of the chloroplast. Energy from ATP and NADPH produced by the light reaction are used to reduce CO_2 . The reactions are catalyzed by enzymes and their sequence was discovered by scientist Calvin. This is an anabolic reaction. The first stable product of the Calvin cycle is glyceraldehyde 3-phosphate (G3P). For the net synthesis of one molecule of G3P, the cycle must take place three times.

The Calvin cycle of photosynthesis can be described in three steps;

- Carboxylation (Carbon fixation)
- Reduction
- Regeneration of carbondioxide acceptor

Carbon fixation

The CO_2 acceptor is a 5 C sugar, Ribulose biphosphate (RuBP). The addition of CO_2 to a RuBP is called carboxylation. The enzyme involved in this reaction is RuBP carboxylase oxygenase or Rubisco.

The first product of RuBP carboxylation is a 6C molecule which is unstable and breaks down immediately into two molecules of 3-phosphoglycerate (3-PGA). This is the first stable product of photosynthesis. The enzyme RuBP carboxylase oxygenase (Rubisco) is present in large amounts in the chloroplast stroma.

Reduction phase

1,3-Bisphosphoglycerate will be reduced to Glyceraldehyde 3- phosphate (G3P) through step by step. Enzyme catalyzed reactions utilizing NADPH and ATP from light reaction. G3P will act as a precursor for carbohydrate synthesis (glucose).

Regeneration of RuBP

RuBP is regenerated by undergoing a series of complex reactions. This process uses energy from ATP generated in light reaction.

Photorespiration

As its name suggests, Rubisco is capable of catalyzing two distinct reactions, acting as both a carboxylase and as an oxygenase.

In the oxygenase reaction of Rubisco uses the same substrate, RUBP, but reacts it with O_2 . The reaction is catalyzed on the same active site as the carboxylation reaction. Thus CO_2 and O_2 are competitive substrates. Therefore CO_2 inhibits the oxygenase and O_2 inhibits the carboxylase reaction.

The oxygenase reaction forms just one molecule of 3-PGA plus a two carbon product, 2-phosphoglycolate which is of no immediate use in the Calvin cycle and in higher concentrations it is toxic for the plant. It therefore has to be processed in a metabolic pathway called photorespiration. The photorespiratory pathway involves enzymes in the chloroplasts, peroxisome and mitochondria. (detail of this pathway is not expected).

Photorespiration is not only energy demanding, but furthermore leads to a net loss of CO_2 . Each time Rubisco reacts with O_2 instead of CO_2 the plants makes 50% less 3-PGA than it would have done if CO_2 had been used. This potentially eliminates the net gain in photosynthetic carbon and loses the productivity.

These two factors result in an increase in photorespiration relative to photosynthesis so that an increasing proportion of carbon is lost as the temperature rises.

The CO_2 required for photosynthesis enters a leaf via stomata. However, stomata are also the main avenues of transpiration. On a hot, dry day, most plants close their stomata in order to conserve water. At the same time O_2 released from the light reactions begins to increase and this leads to further reduction of (CO_2) to (O_2) ratio in the cytosol. These conditions within the leaf favor a wasteful process photorespiration under high temperature, dryness and high light intensities.

Therefore plants developed different way to cope with this situation during the evolution that resulted a most successful solution to concentrate CO_2 around Rubisco provided by C_4 photosynthetic pathway.

The establishment of C_4 photosynthetic pathway includes several biochemical and anatomical modifications that allow plants with this pathway to concentrate CO_2 at the site of Rubisco. Thereby its oxygenase reaction and the following photorespiration are largely repressed in C_4 plants.

In most C_4 plants the CO_2 concentration mechanism is achieved by a division of labor between two distinct specialized leaf cell types, the mesophyll and the bundle sheath cells. Compared to C_3 plants the bundle sheath cells of C_4 plants have expanded physiological functions. This is reflected by the enlargement and higher organelle content of these cells in C_4 species. For the efficient function of the C_4 pathway a close contact between mesophyll and bundle sheath cells are tightly interconnected to each other by high numbers of plasmodesmata. The bundle sheath cells enclose the vascular bundles and are themselves surrounded by the mesophyll cells and this type of leaf anatomy was termed Kranz anatomy.

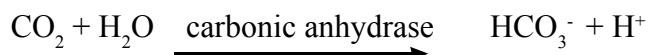
Since Rubisco can operate under high CO_2 concentrations in the bundle sheath cells, it works more efficiently than in C_3 plants. Because of the CO_2 concentration mechanism they can acquire enough CO_2 even when keeping their stomata more closed and minimize the water loss by transpiration.

C_4 pathway of photosynthesis

In the mesophyll cells of C_4 plants CO_2 is converted to bicarbonate by carbonic anhydrase and initially fixed by phosphoenolpyruvate carboxylase using PEP as CO_2 acceptor. The resulting oxaloacetate (OAA) is composed of four carbon atoms, which is the basis for the name of this metabolic pathway. Oxaloacetate is rapidly converted to the more stable C_4 acids malate or aspartate that diffuse to the bundle sheath cells. Here, CO_2 is released by decarboxylating enzymes and the released CO_2 is refixed by Rubisco, which exclusively operates in the bundle sheath cells in C_4 plants.

Chloroplasts found in mesophyll cells are different in anatomy in comparison to chloroplasts of bundle sheath cells.

Since chloroplasts of mesophyll cells carryout only light reaction, they are rich in grana. The grana of mesophyll chloroplasts are large and highly differentiated for light reaction. Bundle sheath chloroplasts possess a very few, less differentiated grana or grana are absent. Moreover, that PSII in the bundle sheath cells are depleted in order to lower oxygen production in these cells.



This PEP carboxylase enzyme is much more efficient than the enzyme of RUBP carboxylase for two reasons.

1. It reacts with bicarbonate (HCO_3^-) rather than with CO_2 . The advantage of this is that there is a 50-fold higher concentration of HCO_3^- than CO_2 in solution in the cytosol.
2. It has no affinity for O_2 .

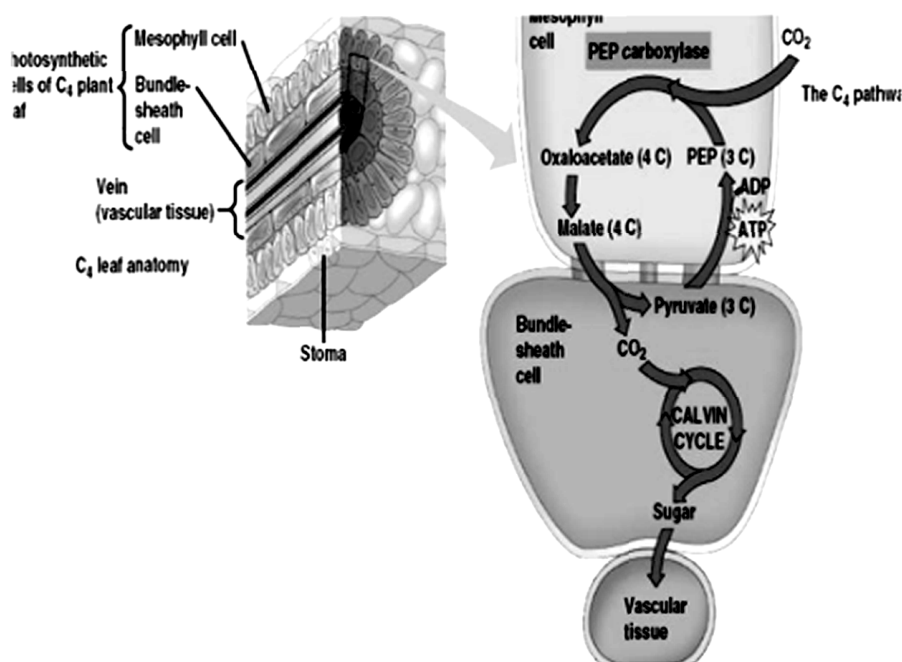


Fig 2.39 : The C₄ Pathway

Significance of the C₄ pathway

- Helps plants to improve the efficiency of CO_2 fixation at lower CO_2 concentrations by preventing the gateways for photorespiration by spatially separating Rubisco.
- In hot-dry climate, it is essential for the stomata to close to prevent water loss through transpiration. This reduces CO_2 intake of particular plants. Therefore, plants in tropical zones or hot climate may suffer from CO_2 deficiency. At lower CO_2 concentrations, C₄ mechanism increases the efficiency of photosynthesis by concentrating CO_2 in the bundle sheath cells.

C4 plants exhibit better water-use efficiency than C3 plants because of the CO_2 concentration mechanism they can acquire enough CO_2 even when keeping their stomata more closed. Thus water loss by transpiration is reduced.

Since Rubisco can operate under high CO_2 concentrations in the bundle sheath cells, it works more efficiently than in C3 plants, consequently C4 plants need less of this enzyme, this leads to a better nitrogen-use efficiency of C4 compared to C3 plants.

Table 2.6: Comparisons of C3 and C4 plants

Characteristics	C3 plants	C4 plants
Representative species	Wheat, rice, barley	Maize, sugarcane, grasses
Temperature optimum for photosynthesis ($^{\circ}\text{C}$)	15-25	50% greater at 35 $^{\circ}\text{C}$
CO_2 fixation	Occurs once	Occurs twice, first in mesophyll cells, second in bundle sheath cells
CO_2 acceptor	5C, RuBP	3C, PEP mesophyll cells 5C RuBP in bundle sheath cells
CO_2 fixing enzyme	Rubisco	PEP carboxylase in mesophyll cells which is very efficient Rubisco in bundle sheath cells, working efficiently under high CO_2 concentration
First product of CO_2 fixation	C3 acid, 3-phosphoglycerate (3-PGA)	4C acid, oxaloacetate (OAA)
Leaf anatomy	Bundle sheath cells, if present, are not green (non photosynthetic), photosynthesis occurs in Mesophyll cells	Kranz anatomy with photosynthesis occurring in both mesophyll cells and bundle sheath cells
Productivity	Yield is usually lower	Yield is usually high

Factors affecting photosynthesis

The rate of photosynthesis is an important factor in crop production. Rate is affected by various factors.

e.g. light intensity, CO_2 concentration, temperature, water, pollutants and inhibitors

The photosynthesis involves a series of reactions. Therefore various factors are involved in it.

Blackman who is the scientist first proposed the idea of principal of limiting factors.

When a chemical process is affected by more than one factor, its rate is limited by the factor which is nearest its minimum value.

e.g. Light intensity

Light Intensity

The rate of photosynthesis increases linearly with increasing light intensity. Gradually the rate of increase falls off as the other factors become limiting. Very high light intensities chlorophyll may bleach and slow down photosynthesis. However, plants exposed to such conditions are usually protected by devices such as thick cuticles, hairy leaves.

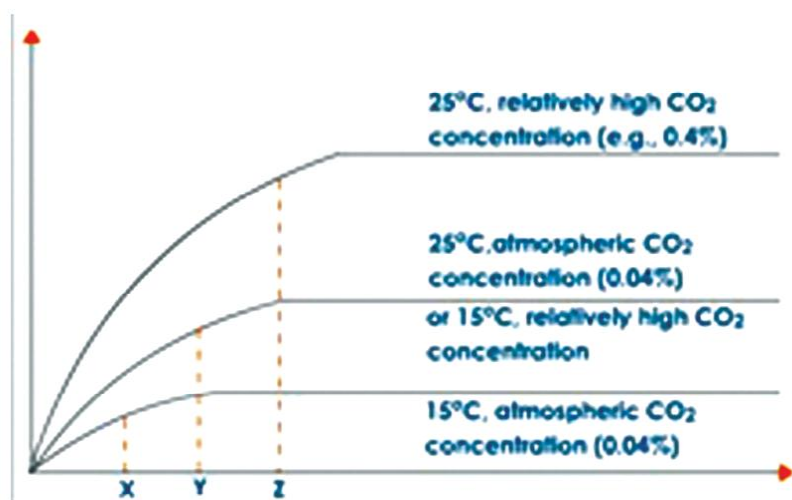


Fig 2.40: Rate of photosynthesis with light intensity at different temperatures

Under normal conditions, CO_2 is the major limiting factor in photosynthesis. Increase in photosynthetic rate is achieved by increasing CO_2 concentration. For example some greenhouse crops such as tomatoes are grown in CO_2 enriched atmosphere.

Cellular respiration as a process of obtaining energy

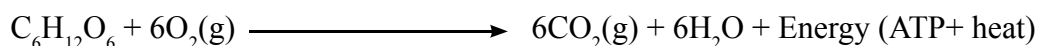
Cellular respiration is the process by which chemical energy in organic molecules such as carbohydrates is released by stepwise oxidative process, catalyzed by enzymes and made available in living cells in the form of ATP. Cellular respiration is divided as

- a) aerobic respiration
- b) anaerobic respiration

Aerobic respiration:-

The process of synthesize ATP from the respiratory substrates such as glucose in the presence of molecular oxygen (O_2) known as aerobic respiration. Glucose is found to be the major respiratory substrate in living cells.

The aerobic respiration of glucose molecules can be represented by the following balanced chemical equation.



This process consist three main steps. They are;

- a) Glycolysis
- b) Pyruvate oxidation and citric acid cycle (Kreb's cycle)
- c) oxidative phosphorylation (Electron transport chain)

Glycolysis

It takes place in the cytosol of the cell, because all enzymes that catalyze reactions of the glycolysis are found in the cytosol of the cell. This process does not depend on O_2 . During the above process a six carbon (6C) glucose molecule is broken down step by step into two three-carbon (3C) pyruvate molecules.

Two ATP molecules are used up to initiate the process.

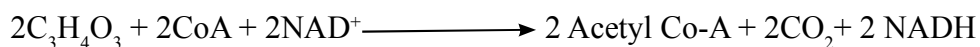
Four hydrogen molecules and electrons released from glucose breakdown reduce two NAD^+ and produce two NADH. At the end of glycolysis there will be four ATP molecules produced. Since two ATP molecules were used up for the initiative step, the net yield will be two ATP molecules.

Only when O_2 is present, the pyruvate molecules will enter the mitochondria and further steps will take place.

Oxidation of Pyruvate/ Link reaction

These two pyruvate molecules enter mitochondrion by active transport through the membrane. In the matrix of mitochondria, Pyruvate is converted to acetyl group by releasing two CO_2 molecules. Then this acetyl group combines with co-enzyme A to

produce Acetyl co-A. In this reaction two NAD^+ is converted to two NADH molecules. Therefore this step can be represented as follows.



Oxidation of pyruvate is a linking reaction of glycolysis and citric acid cycle.

Acetyl Co-A will feed its acetyl group for citric acid cycle.

Citric acid cycle

This takes place in the matrix of mitochondria using specific enzymes. As the first product of this cyclic pathway is citric acid, it is known as citric acid cycle. The pathway was discovered by a German-British scientist Hans Krebs. Hence, it is named as Krebs' cycle. Citric acid contains three carboxylic acid groups. This cycle is also known as Tricarboxylic Acid cycle or TCA cycle. In the citric acid cycle 4 C compound oxaloacetate combines with 2 C compound acetyl Co -A to form 6 C compound, citric acid. Then citric acid undergoes a series of reactions to regenerate oxaloacetate by releasing two CO_2 molecules by decarboxylation reaction. One ATP molecule is produced by substrate level phosphorylation. One FADH_2 and three NADH will be generated as a result of oxidation reactions. These are the products of a single acetyl group led into citric acid cycle and hence these numbers should double when the yield for a glucose molecule is considered.

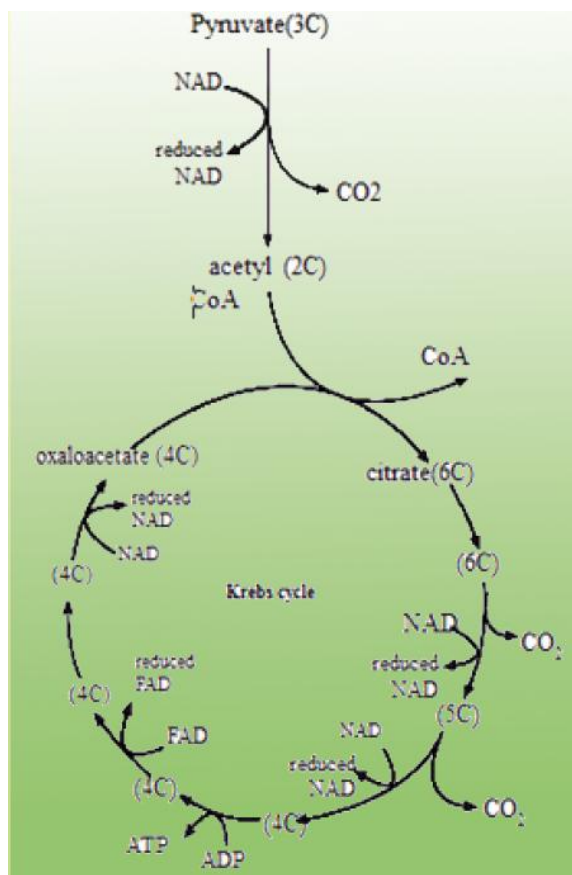


Fig 2.41: Krebs cycle (mechanism is not necessary for the examination)

Electron transport chain

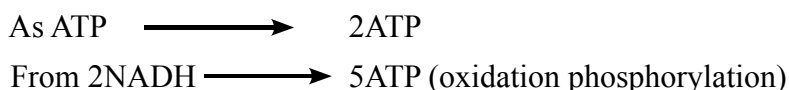
This step is taken place across the inner membrane (cristae) of mitochondria. The folding of cristae increases surface area for oxidative phosphorylation. NADH and FADH_2 products in the early stages of aerobic respiration are oxidized by transferring electrons, through the electron transport chain and finally to molecular oxygen (O_2). The electron transport chain is located in the inner membrane of mitochondrion and composed of series of protein and non-protein molecules involving in the movement of electrons and protons across cristae. Therefore, the Molecular oxygen(O_2) is the final electron acceptor in aerobic respiration. In the electron transport chain, ATP is synthesized by oxidative phosphorylation.

In this electron transport chain, energy is released progressively from NADH and FADH_2 and that energy is used to synthesize ATP. When one molecule of NADH is oxidized in the electron transport chain, 2.5 molecules of ATP in average are generated due to oxidative phosphorylation. When one molecule of FADH_2 is oxidized 1.5 molecules of ATP in average are produced due to oxidative phosphorylation. Total number of ATP that is produced in this step is 28.

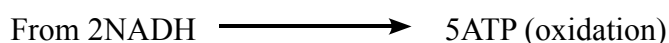
This is true in the active cells such as liver cells and cardiac muscle cells but not in other cells where two ATP produced in glycolysis is used to transport 2NADH from cytosol to mitochondrial matrix. In those cells total number of ATP produced by one molecule of glucose is $(32-2) = 30$ ATP.

Total number of ATP molecules produced from one molecule of glucose, during aerobic respiration.

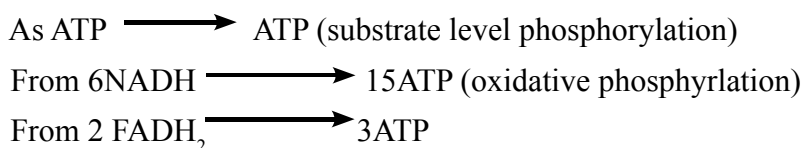
In glycolysis;



In pyruvate oxidation;



In Citric acid cycle;



$$\therefore \text{Total Number of ATP} = 32 \text{ ATP}$$

Anaerobic respiration

Anaerobic respiration is breaking down of glucose in the absence of molecular oxygen which is regulated by enzymes of the cells occurring in cytosol. In the absence of molecular oxygen pyruvate molecules cannot be broken down further. ATP generated is utilized to fulfill energy requirements. However, NADH produced during glycolysis cannot be utilized. Therefore, since NAD^+ is limited it is essential for the cell to recycle NADH to enhance the availability of NAD^+ other than anaerobic respiration

fermentation is a method of production of ATP in the absence of O_2 . There are many types of fermentation, differing from end products formed by pyruvate. The two common types are;

1. Ethyl alcohol fermentation
2. Lactic acid fermentation

Ethyl alcohol fermentation

- Like in aerobic respiration, the first step of this is also Glycolysis.
- Therefore one molecule of glucose is converted to 2 molecules of pyruvate giving 2 molecules of ATP and two molecules of NADH
- Then this pyruvate involve in two steps. In the 1st step pyruvate is converted in to Acetyldehyde, releasing a molecule of CO_2 .
- In the second step acetyldehyde is reduced to ethanol using NADH that is produced in Glycolysis.
- Therefore final hydrogen acceptor in ethyl alcohol fermentation is acetylaldehyde (organic compound)
- Many bacteria carry out ethyl alcohol fermentation. The most common organism which carries out ethyl alcohol fermentation is yeast.

Lactic acid fermentation

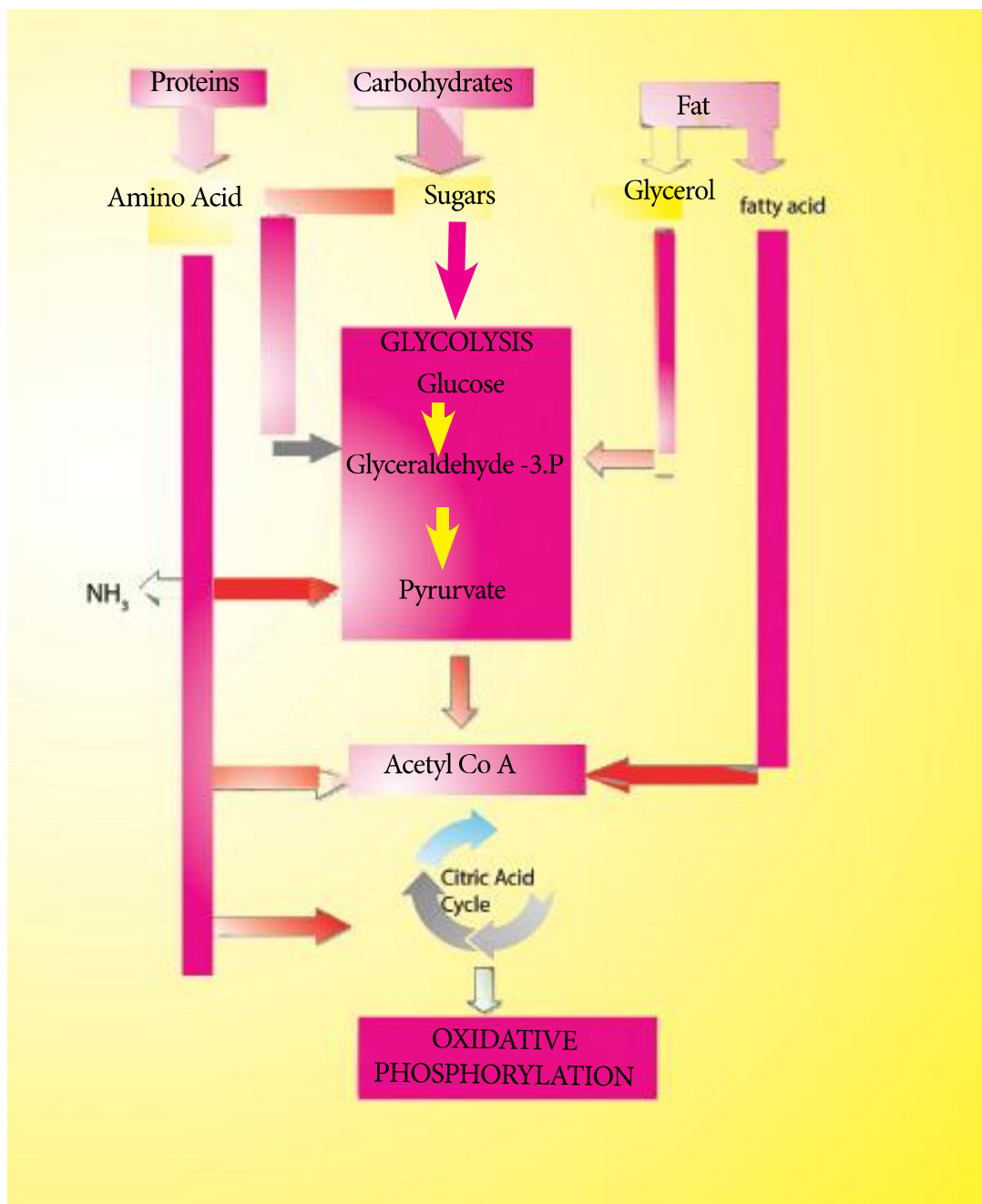
- As in ethyl alcohol fermentation, Glycolysis takes place as the first step of lactic acid fermentation.
- Therefore one molecule of glucose produces two molecules of pyruvate, two molecules of ATP and two molecules of NADH.
- Then pyruvate is reduced directly by NADH for lactic acid as an end product with no release of CO_2 . Therefore final H acceptor is also organic compounds.
- Certain fungi and bacteria carryout lactic acid fermentation but the most common organisms are lactic acid bacteria involved in formation of yoghurt and curd.

Respiratory quotient

It is the ratio of CO_2 evolved and the volume of O_2 consumed in a given time for the respiratory substrate.

$$RQ = V CO_2 / V O_2$$

RQ of respiration of carbohydrates, fats and proteins are 1.0, 0.7 and 0.8 respectively.

Use of proteins, Carbohydrates, fats in respiration

03**Evolution and Diversity of Organisms****The theories of origin of life and natural selection to analyze the process of evolution of life****Origin of life on earth****Condition of earth before life**

Earth and the other planets of the solar system were formed about 4.6 billion years ago. At the beginning of the solar system, planet Earth was being bombarded by chunks of rocks and ice.

The first atmosphere was probably thick with water vapour, along with various compounds released by volcanic eruptions, including nitrogen and its oxides, carbon dioxide, methane, ammonia, hydrogen and hydrogen sulphide. The neutral atmosphere then turned to be a reducing one. The first atmosphere had little oxygen. Later earth was cooled down and the water vapour condensed into the ocean. Some of the hydrogen quickly escaped into the space. Volcanic eruptions, lightening, extreme UV radiation, hydro thermal vents and alkaline vents along with the Earth's reducing atmosphere favored the synthesis of organic molecules essential for the origin of life. These simple organic molecules then polymerized to form macromolecules such as proteins and nucleic acids. Further, the formation of self-replicating organic molecules made life possible on earth.

Evolution of Biological Diversity**1. Biochemical evolution**

Direct evidence for life on early earth comes from fossils of micro-organisms that are about 3.5 billion years old. Observations and experiments in chemistry, geology and physics have provided evidence, for the appearance of the first living cells. The theory of the biochemical evolution arose from the hypothesis based on chemical and physical processes on early earth. The emerging force of natural selection could have produced the first cells through a sequence of four main stages.

1. Atmospheric conditions of early earth facilitated the abiotic synthesis of small organic molecules such as amino acids and nitrogenous bases from inorganic molecules

2. Polymerization of the above small organic molecules leads to the formation of organic macromolecules.

- a. Amino acids $\xrightarrow{\text{polymerization}}$ proteins
- b. N-base + sugar + phosphate $\longrightarrow$ Nucleic acids

3. Organic macromolecules were packed into membranes, to produce protocells

4. Nucleic acids gained self replicating capability, which made inheritance possible for the cells.

2. Origin of protocell

Haldane suggested that the early oceans were a solution of organic molecules “primitive soup” in which life arose. Recent studies related to the volcanic-atmosphere and alkaline vents show the abiotic synthesis of organic molecules. Another source of organic molecules may have been meteorites. RNA accumulated into lipid bound vesicles and formed “protocells” which exhibited enzyme catalyzed activities and were able to grow, replicate and evolve. The early genes and enzymes would have been RNA which enabled replication of RNA. Other molecules that were in the primitive soup were also collected in the protocell. Growth occurred by addition of lipids to the membrane by collision of micelles. When the protocell becomes too large, it divided to form two protocells with RNA included.

3. Origin of photosynthetic organisms

Fossils of the first photosynthetic organism, today’s cyanobacteria, originated before 2.7 billion years ago. As a result of photosynthesis iron (Fe^{2+}) ion were oxidized. Once all of the dissolved iron has precipitated, additional O_2 dissolved in the water until the water bodies became saturated with O_2 . The increase of photosynthetic bacteria contributed to the increment of the amount of atmospheric oxygen which had accelerated the origin of chloroplast.

4. The origin of first eukaryote

The fossils of the first eukaryotic organisms were estimated as from about 1.8 billion years ago. These eukaryotic single cellular organisms later evolved in to multicellular organisms. The appearance of structurally complex eukaryotic cells sparked the evolution of greater morphological diversity than was possible for the simple prokaryotic cells. After the first eukaryotes appeared, a great range of unicellular forms evolved. It gave rise to diversity of some single-celled eukaryotes which evolved in to multicellular forms, such as the varieties of algae, plants, fungi and animals. Fossils of the oldest known protists similar to small red algae were dated as 1.2 billion years ago.

5. Diversification of Eukaryotes

Many present day animal phyla appeared in the early Cambrian period. Several animal groups which include, porifera, sponges, cnidarians (Sea anemones and their relatives) and molluscs appeared in the late Proterozoic. According to the DNA analysis, sponges evolved 700 million years ago. Ancestors of arthropods, chordates and other animal phyla originated 670 million years ago. The first food chains on earth appeared when animals started to depend on algae or plants as consumers and with the arrival of many groups of animals, functioning food webs began to appear. Colonization of land by fungi, plants and animals began after about 500 million years ago. Plants that colonized land possess vascular systems to transport water and minerals and water proof coating of wax to prevent the water loss. With the emergence of large trees, differentiation as roots, stems and leaves began and diversified since 40 million years ago. Plants and fungi colonized the land together by interacting with each other. Arthropods (insects and spiders) were the first group of animals to colonize the land. The earliest tetrapods formed about 365 million years ago which were evolved from lobed-finned fish. The divergence of human lineage from other primates was initiated 6-7 million years ago. The origin of the human species took place 195,000 years ago.

Geological eons and eras of evolution

- Eons: Hadean, Archaean, Proterozoic, Phanerozoic
- Eras: Eon Phanerozoic covers the 3 eras, Palaeozoic, Mesozoic, Cenozoic

1. Hadean eon

- Origin of Earth

2. Archaean eon

- Oldest known rocks on Earth's surface
- Oldest fossils of cells (prokaryotes) appeared
- Concentration of atmospheric oxygen begins to increase

3. Proterozoic eon

- Diverse algae and soft-bodied invertebrate animals appeared
- Oldest fossils of eukaryotic cells appeared

Eon Phanerozoic covers the 3 eras; Palaeozoic, Mesozoic, Cenozoic

1. Palaeozoic era

- Sudden increase in diversity of many animal phyla
- Marine algae becomes abundant; colonization of land by diverse fungi, plants, and animals
- Diversification of vascular plants
- Diversification of bony fishes, first tetrapods and insects appeared
- Amphibians dominated
- Extensive forests of vascular plants

- First seed plants appeared
- Origin and radiation of reptiles
- Origin of most present-day groups of insects
- Extinction of many marine and terrestrial organisms
- Diversification of early vascular plants

2. Mesozoic era

- Cone-bearing plants (gymnosperms) dominated
- Dinosaurs evolved, radiated
- Origin of mammals
- Gymnosperms continued as dominant plants, dinosaurs dominated, abundant and diverse
- Flowering plants (angiosperms) appeared and diversified, many organisms including dinosaurs become extinct

3. Cenozoic era

- Major radiation of mammals, birds, and pollinating insects
- Dominance of angiosperm increased and their radiation continued, radiation of most present day mammalian orders
- Origins of many primate groups, continued radiation of mammals and angiosperms, earliest direct human ancestors
- Appearance of bipedal human ancestors
- Origin of genus Homo

Theories of evolution

Evolution can be defined as a change in the genetic composition of a population from generation to generation (descent with modification) over a long period of time. This may take millions of years.

Theories of evolution are

- Theory of Lamarck.
- Darwin - Wallace theory (Theory of Natural selection)
- Neo Darwinism

Theory of Lamarck

Lamarck published his hypothesis in 1809. He explained his hypothesis using two principles.

1. Use and disuse
2. Inheritance of acquired characteristics

1. Use and disuse - The parts of the body that are used extensively become larger and stronger. If not used, they deteriorate.
e.g.- Giraffe stretching its neck to reach leaves on higher branches.
2. Inheritance of acquired characteristics – Organism acquired adaptation during their life time according to the needs of environment. Offspring is better adapted to live in that environment e.g. long muscular neck of the giraffe had evolved over many generations as giraffes stretch their necks even higher

Darwin - Wallace theory (Theory of natural selection)

Darwin observed two phenomena from the environment. His observations were;

The populations of a species vary in characteristics among their inheritance traits.

Each species produces more offspring than their environment could accommodate.

The above observations were interpreted by Charles Darwin as,

Certain traits of a population which are capable of exhibiting qualities for better survival and their reproduction can produce more offspring.

Variation in abilities for survival and reproduction among a population may enhance the abundance of favorable characteristics in that population.

Some favorable characteristics for survival and reproduction are;

- Escaping from predators - defense
- Tolerating physical conditions – stress conditions
- Obtaining food
- Resistance against disease
- Fertilizing probability
- Number of offspring produced

Process of natural selection

- Over production
- Variation
- Competition and survival of the fittest
- Natural selection of favourable traits

Neo-Darwinism

Neo-Darwinism generally denotes the integration of Charles Darwin's theory of natural selection, Mendelian genetics as the basis for biological inheritance and knowledge of population genetics.

Hierarchy of taxa on scientific basis

Methods of artificial and natural classification

Arrangement of organisms into groups on the basis of the common characteristics is called classification. Taxonomy is the scientific study on classification, identification, nomenclature and description. This includes placing groups of organisms in a hierarchical sequence.

Two methods of classification

(1) Artificial classification - grouping is based on a few pre selected unifying characters.

- The characters are selected first according to convenience and organisms are grouped based on the selected criteria
- Evolutionary relationships are ignored
- Only system used before 18th century
- Easy to use, easy to expand by adding more groups

e.g. Plants can be classified as cereals, ornamental plants, medicinal plants, poisonous plants etc. Animals can be classified as two legged, four legged, six legged, eight legged etc.

(2) Natural classification - grouping based on true relationships.

- Represent s evolutionary relationships based on phylogeny- evolutionary history of a species or groups of species
- Systems developed after then study of evolution.
- Based on many characteristics.

Characteristics used can be morphological, anatomical, cytological or molecular biological such as DNA and RNA base sequences

e.g. Plants can be classified into phyla; Bryophyta, Lycophyta, Pterophyta, Cycadophyta, Coniferophyta and Anthophyta etc. Animals can be classified into Cnidaria, Platyhelminthes etc.

History of classification

The early classification systems were all artificial systems and were mostly based on human uses.

Aristotle was the first to classify organisms scientifically. He divided organisms into plants and animals. Animals were further classified according to criteria such as mode of locomotion, reproduction and presence or absence of red blood cells. Aristotle's pupil Theophrastus classified plants according to habit. e.g. trees, shrubs and herbs, and according to lifespan e.g. annuals, biennials and perennials.

Up to the time of Linnaeus scientists have used many different methods for naming of organisms. Carolus Linnaeus(1753), Swedish botanist, introduced binomial nomenclature and also classified about 6,000 plants into a hierarchical order of taxa, classification level such as; Species, genus, order, and class. His classification of flowering plants was based on the number of stamens and styles of flower. He identified two kingdoms of organisms; plants and animals.

With the discovery of the microorganisms the scientists understood that there were organisms which could not be assigned into either plants or animals. To get over this difficulty Ernest Haeckel (1866) introduced a third kingdom: Protista. He also introduced the taxon Phylum and classified many organisms.

With the discovery of the electron microscope biologists identified prokaryotic and eukaryotic cellular organization. Robert H Whittaker (1969) introduced the five kingdom system of biological classification; Monera, Protista, Fungi, Plantae and Animalia. His classification was based on the nature of cellular organization, unicellular or multicellular and mode of nutrition.

With the acceptance of Darwin's theory of the evolution and unitary origin of life, taxonomists began to use natural systems with interpreting evolutionary relationships. With the recent advancement of molecular biology and the use of molecular methods in studying evolutionary relationships it became apparent that in the very early evolution some prokaryotes differ as much from each other as do from eukaryotes. Such difficulties have lead biologists to adopt three Domain system of classification. The three domains are Bacteria, Archea and Eukarya, which are taxonomic ranks higher than the Kingdom. Carl Woese (1977) introduced this three domain system.,

In this tree of life the first major split in the history of life occurred when bacteria diverged from others. Eukarya and Archea are mostly related to each other than bacteria.

Present system of classification and its basis

The present system of classification is mainly based on the rapid advance of molecular biology and the new information on the evolutionary relationships of organisms.

- the sequence of bases of DNA of important genes
- the sequences of bases of DNA of mitochondria and chloroplasts
- the base sequence of ribosomal RNA
- the sequence of amino acids in common proteins
- the molecular structure of cellular components

are used as important taxonomic criteria in modern systematics.

However, the kingdom Protista is not a natural group. It is an artificial group including organisms which have different evolutionary origins.

Viruses do not have cellular organization, and therefore do not belong to any of the kingdoms. They are also an artificial group considered separately.

Hierarchy of Taxa from Domains to Species

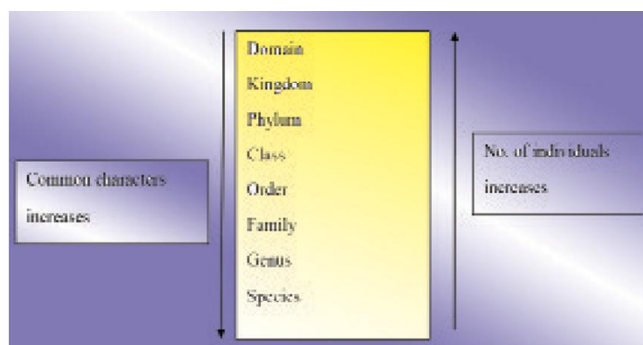
The taxonomic unit at any level/ rank of the hierarchy is called a taxon (plural-taxa). Each taxon has a rank and a name.

e.g. *Panthera* is a taxon at the Genus level/ rank

Mammalia is a taxon at the Class level/ rank

Under the hierarchical system there are levels/ ranks of taxa. Each Domain is divided into kingdoms. A Kingdom is divided into phyla (singular phylum), phylum into classes .etc. Many of these categories may also be subdivided.

e.g. Super class, Sub-family, Subspecies, etc.



From domain to species, the number of shared characteristics among the members in the taxa increases. From species to domain, the number of individuals in the taxon increases.

Biological definition of a species

Species is a group of organisms who shares similar characteristics and has the ability to interbreed and produce viable and fertile offspring.

Other definitions of species

- Morphological species concept- use of morphological criteria to distinguish species such as body shape and other structural features

- Ecological species concept—defines species in terms of its ecological niche and the sum of how members of the species interact with the non living and the living components of their environment
- Phylogenetic species concept – defines the species as the smallest group of individuals that share a common ancestor.

Binomial nomenclature

In classification, use of common names for organisms, causes confusion. More over some common names do not actually reflect the kind of organism they signify.

- e.g. Jelly fish (a Cnidarian)
Cray fish (a Crustacian)
Silver fish (an insect)
Star fish (an Echinoderm)

Further, a given organism has different names in different languages. Carolous Linnaeus (1707-1778) proposed a binomial system of nomenclature of species, which was accepted worldwide to avoid ambiguity.

According to the binomial nomenclature the name of an organism has two parts:

First is the generic name, to which the species belongs and the second is a specific epithet, the unique for each species within the genus. Generic name is usually a noun and the specific epithet an adjective describing a particular feature.

e.g. *Homo sapiens*- *Homo* means man, *sapiens* means intelligent

Related species have the same generic name with different specific epithets.

e.g. *Dipterocarpus zeylanicus* and *Dipterocarpus grandiflorus*

Dipterocarpus zeylanicus means fruit with two wings, and endemic to Sri Lanka.

Dipterocarpus grandiflorus means fruit with two wings and having large flowers.

International codes of Binominal nomenclature

Biologists have adopted sets of rules or Codes of nomenclature. These codes are slightly different for plants, animals, bacteria and viruses. Some of the important rules for naming plants, fungi, bacteria and animals are as follows.

- Two species of organisms cannot have the same name.
- Each species has a generic name and a specific epithet, both together forming the species name or scientific name.
- The Name should be made up of Latinized words written in the Roman script.
- It should be underlined when hand written and italicized when printed.
- The first letter of the generic name must be capitalized the and specific epithet must be in simple letters.

In scientific writing, the name of the author who gave the name is indicated by a capital letter, an abbreviation or full word at the end of the name, which is not Latinized.

e.g. *Cocos nucifera* L., (L for Linnaeus).

A third word can be used to represent a subspecies or a variety, example *Panthera parduskotiya* (Sri Lankan leopard).

- **Use of keys**
 - Used to group organisms and identify them
 - Keys do not show the evolutionary relationships
 - The Commonly used key is the dichotomous key
 - Some examples are given below

Example 1: Silverfish, Butterfly, House fly, Beetle

1. Possesses wings (2)
Do not possess wings Silverfish
2. Possess two pairs of wings (3)
Do not possess two pairs of wings.....Housefly
3. Possesses a proboscisButterfly
Do not possess a proboscisBeetle

Example 2: Snake, Earthworm, Frog, Sea anemone, Butterfly

1. Radially symmetrical bodySea anemone
Not having a radially symmetrical body(2)
2. Possess legs.....(3)
Do not possess legs(4)
3. Wings present..... Butterfly
Wings absent..... Frog
- 4 Body covered by scales Snake
Body is not covered by scales Earthworm

Domains

There are three domains. They are;

- a) Domain – Bacteria- consists of one kingdom. Kingdom - Bacteria
- b) Domain –Archaea-consists of one kingdom. Kingdom - Archaeobacteria
- c) Domain –Eukarya-consists of four kingdoms.

Kingdom - Protista
Kingdom - Fungi
Kingdom - Plantae
Kingdom - Animalia

The diversity of organisms within the Domain Bacteria

Key characteristics of Domain Bacteria

- They are prokaryotic
- They are unicellular, colonial, filamentous
- Most of them are found in size between 0.5 to 5µm
- Well adapted to most of the 'normal' habitats (both land and water)
- Most of them contain peptidoglycan in their cell walls
- According to the amount of peptidoglycan present in the cell wall they are classified as Gram positive and Gram negative bacteria
- Most of their cell walls are surrounded by a sticky layer of polysaccharides or proteins called capsule
- Most of them have flagella for motility. Bacterial flagellum differs from eukaryotic flagellum as they are not covered by a plasma membrane and absence of 9+2 structure of microtubules.
- Possess diverse nutritional modes-Autotrophs, heterotrophs
- Posses diverse metabolic modes- obligate aerobes, obligate anaerobes, facultative anaerobes, etc.
- Some are capable of performing nitrogen fixation- e.g. *Rhizobium sp.*, some cyanobacteria
- Rapid reproduction by binary fission. Some perform conjugation as a sexual method.
- Certain bacteria use bacterial chlorophyll as a photosynthetic pigment.

Key Characteristics of Cyanobacteria

- Prokaryotic organisms
- Photosynthetic
- Most are unicellular and oxygen generating and solitary. But some are linked to form filaments or colonies sheathed in mucous
- Some have the ability of fixing atmospheric nitrogen

Key characteristics of Domain Archaea

- They are prokaryotic and unicellular.
- They lack peptidoglycan in their cell walls which are made up of proteins and polysaccharides
- The size of most of them is between 0.5-5 μm
- They include extreme halophiles and extreme thermophiles
- Some Archaeabacteria live in more moderate environments-Methanogens
- Other species inhabits the anaerobic guts of cattle, termites and other herbivores

Key characteristics of Domain Eukarya

- They are Eukaryotic
- Vary in size
- Most of them are multicellular
- Habitats are diverse
- Diverse in nutrition
- Mostly aerobes
- Most of them exhibit sexual reproduction (some protists are only known to reproduce asexually)

Table 3.1: A comparison of the three domains of life

	Characteristic	Bacteria	Archea	Eukarya
1	Cellular organization	Prokaryotic	Prokaryotic	Eukaryotic
2	Cell wall composition	Peptidoglycan	Proteins and polysaccharides (lack peptidoglycan)	Cellulose, Hemicellulose, Pectin and Chitin
3	Membrane lipids	Unbranched hydrocarbons	Some branched hydrocarbons	Unbranched hydrocarbons
4	Genetic Composition			
	Histones associated with DNA	Absent	Present in some species	Present
	Circular chromosomes	Present	Present	Absent
	Introns in genes	Very rare	Present in some genes	Present in many genes

5	Protein synthesis			
	RNA polymerase	One kind	Several kinds	Several kinds
	Initiator amino acids for protein synthesis	Formyl-methionine	Methionine	Methionine
6	Response to antibiotics Streptomycin and Chloramphenicol	Growth inhibited	Growth not inhibited	Growth not inhibited
7	Growth at temperatures > 100°C	No	Some species	No
8	Habitats	Diverse habitats	extreme environmental conditions-volcanic pits/ hot springs/ salt marshes etc.	Diverse habitats
9	Examples	Bacteria, cyanobacteria; <i>Nostoc</i> , <i>Anabaena</i> ,, <i>Escherichia coli</i> , <i>Salmonella typhi</i>	Archaeobacteria; <i>Methanococcus</i> <i>Thermococcus</i> ., <i>Helobacteria</i> .	Protists fungis plants and animals

The diversity of organisms within the kingdom Protista

Key characteristics of Kingdom Protista

- Most of them are unicellular, although there are some colonial and multi cellular species
- It is a polyphyletic group (originated from more than one ancestor) and an artificial group in classification.
- Found in freshwater, marine and damp soil, some are symbionts.
- Unicellular, colonial or multicellular.
- Some are photoautotrophs, some are heterotrophs and some are mixotrophs (combination of photoautotrophic and heterotrophic nutrition).

Euglena

- Unicellular, lack cell wall and pellicle present.
- Chloroplasts are present.
- They have one or two flagella
- They have eye spot
- Contractile vacuole is present
- They have a pocket at one end of the cell from which one or two flagella emerged.

***Paramecium***

- Habitat is freshwater
- Lack cell wall but pellicle is present, unicellular
- Cilia may completely cover the cell surface
- They have two types of nuclei- mega nucleus and micronucleus
- Contractile and food vacuoles are present
- Oral groove is present

***Amoeba***

- Aquatic (marine and freshwater) forms are free living others are parasitic.
- Lack cell walls, unicellular organisms
- They form pseudopodia which are used to locomote and feed
- They do not have definite shape.
- Food vacuoles are present

***Ulva***

- Macroscopic marine forms.
- Cell wall present
- Multicellular thallus differentiated into leaf like blades and root like holdfast.
- Green in colour (green algae)

***Gelidium***

- Marine.
- Cell walls present
- Multicellular thallus with hold fast.
- It is greenish red in colour (red algae)

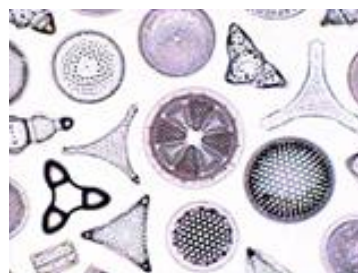
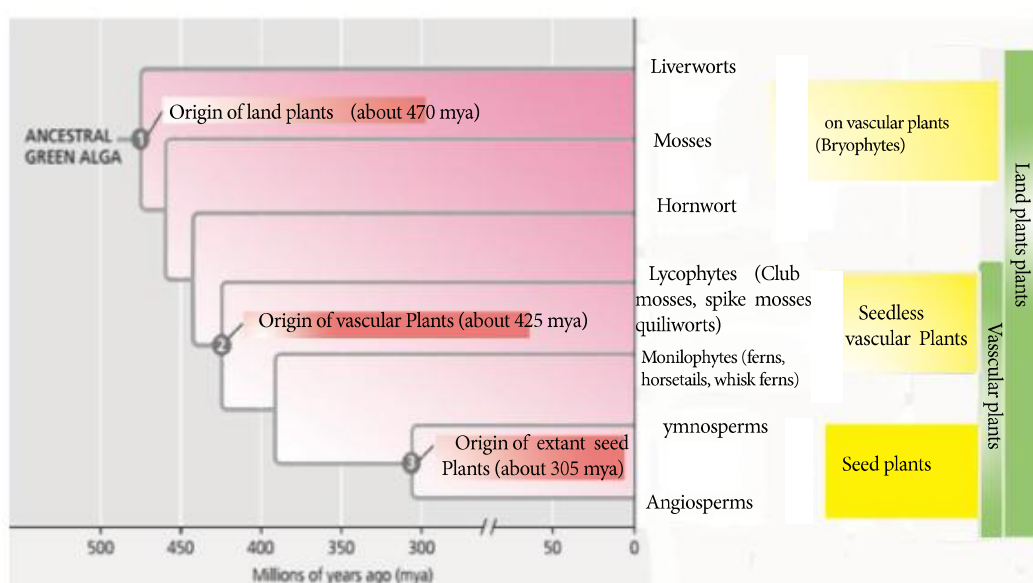


Sargassum

- Marine
- Comparatively larger and complex
- Thallus is plant-like; it consists of a root like holdfast, stem like stipe and leaf like blade.
- Multi cellular, thallus is supported by gas filled bulb shape floats.
- Appear in olive green or brown colour (brown algae)

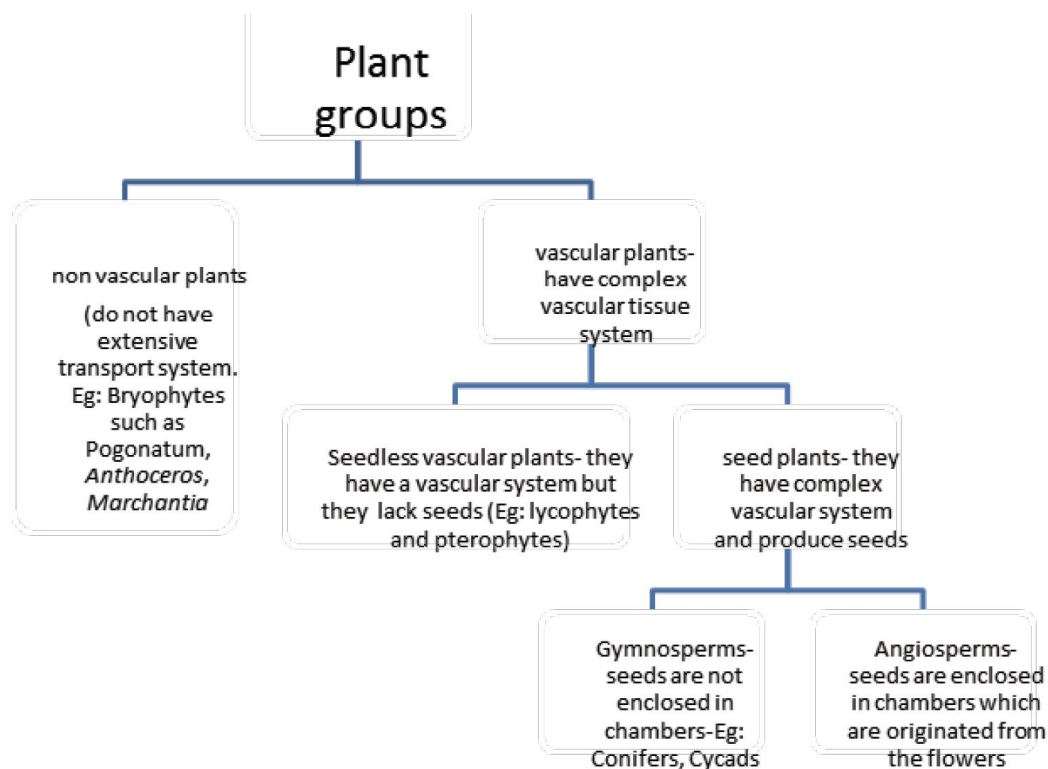
**Diatoms**

- It is aquatic (fresh water and marine)
- Unicellular, having glass like, wall consists of two parts that overlap (presence of silica)
- Highly diverse group regarding the shape and markings in the surface
- Golden brown in colour (golden brown algae)

**The diversity of organisms within the kingdom Plantae****Kingdom Plantae****Evolutionary relationships among major groups of plants****Fig 5.1 Evolutionary relationship of major groups of plants**

It is believed that members of the kingdom Plantae were evolved from a group of chlorophytes/ green algae. Most of them are terrestrial organisms. Chlorophyte algae lack key traits of land plants, walled spores produced in sporangia, multicellular gametangia, dependent embryo and apical meristem. They evolved in the terrestrial environment.

Plant groups can be distinguished based on the presence or absence of an extensive system of vascular tissue. Based on that there are two major groups of plants the can be seen; they are Vascular plants and Non-vascular plants.



Diversification of Kingdom Plantae

- Non-vascular plants
 - Phylum Bryophyta - Mosses- *Pogonatum*
 - Phylum Hepatophyta- *Marchantia*
 - Phylum- Anthocerophyta- *Anthoceros*
- Vascular seedless plants
 - Phylum – Lycopphyta- *Selaginella*
 - Phylum – Pterophyta- *Nephrolepis*
- Vascular seed plants

- Gymnosperms- Phylum – Cycadophyta (*Cycas sp.*),
Phylum- Coniferophyta (*Pinus*), Phylum – Gnetophyta (*Gnetum*)
- Angiosperms- Phylum – Anthophyta (all flowering plants)

Non-vascular plants

One way to distinguish a group of plants is to see whether or not they have an extensive system of vascular tissue that transport water and nutrients throughout the plant body. Most present day plants do have a complex vascular tissue system and therefore, are called vascular plants. Plants that do not have an extensive transport system are described as non vascular plants. Non vascular plants are informally named as Bryophytes.

e.g. *Marchantia*, *Pogonatum*, *Antheroceros*

Bryophytes share some derived traits with vascular plants, but lack many innovations of vascular plants such as presence of true stems, roots and leaves.

Diversity of Bryophytes

Phylum- Hepatophyta e.g. *Marchantia* (liverworts).

Phylum Anthocerotophyta e.g. *Anthoceros* (hornworts)

Phylum Bryophyta e.g. *Pogonatum* (mosses)

Characteristic features of phylum Bryophyta

e.g. *Pogonatum*

- Especially common in moist terrestrial places.
- Haploid gametophyte is dominant stage of the life cycle, photosynthetic and independent.
- Gametophytes are differentiated into ‘leaves’, ‘stems’ and rhizoids. They have no vascular tissues. Archegonia and antheridia are typically carried on separate female and male gametophyte. Therefore gametophyte is dioecious.
- Male plant produces flagellated sperm which can swim through a film of water for fertilization.
- Sporophytes are usually green and photosynthetic when young. However, they are not independent. They attach to their parental gametophytes and absorb nutrients and water from the female gametophyte.
- Sporophytes have specialized pores called stomata which are also found in all vascular plants.
- They are homosporous.

Vascular plants cover about 93% of the existing plant species. They can be further divided into two groups.

1. Seedless vascular plants
2. Seed plants

Seedless vascular plants

Seedless vascular plants lack seeds and disperse by means of spores. They are categorized into two groups.

1. Lycophytes
2. Pterophytes

Even though, both pterophytes and lycophytes are seedless plants, pterophytes share a more recent common ancestor with seed plants.

Fossils and living seedless vascular plants provide evidence for plant evolution during Devonian and Carboniferous periods. The ancestors of vascular plants already have had some derived traits of modern vascular plants; however, they lack roots and some other adaptations.

Fossils suggest that, the ancestors of vascular plants had gametophyte and sporophytes that were about equal in size. However among the living vascular plants sporophyte generation is large and more complex. For example in ferns, leafy plants are the sporophytes



Significant features of seedless vascular plants;

1. Transportation through Xylem and Phloem

Vascular plants have two types of vascular tissues; Xylem and Phloem

Xylem consists of tracheids, fibers and parenchyma cells- conducts water and minerals.

Cell walls of tracheids and fibers are strengthened by the polymer lignin. These tissues permit plants to grow tall. This may facilitate them to obtain a high amount of light for photosynthesis and ease the spore dispersal.

Phloem- this tissue has cells arranged in tubes. They distribute sugars, amino acids and other organic products among different parts of the plant.

2. Evolution of roots

Roots are organs that absorb water and nutrients from the soil. They anchor the plants and allow the shoot system to grow taller. They are to replace the rhizoids seen in bryophytes. Root tissues of living plants resemble stem tissues of the early vascular plants preserved in fossils.

3. Evolution of leaves- There are two types of leaves. They are microphylls and megaphylls. Microphylls are single veined and smaller in size while megaphylls are large, flattened with branched veins.

Leaves with branched vascular tissues increase the surface area for efficient photosynthesis (megaphylls).

Sporophylls and spore variations

Modified leaves that bear sporangia are known as sporophylls. Most seedless vascular plant species produce one type of sporangium and one type of spores. Therefore, they are known as homosporous.

Some plant species produce two types of sporangia and produce two kinds of spores called mega spores and microspores. This condition is known as heterosporous. Mega spores develop into female gametophyte while microspores develop into male gametophyte.

Phylum Lycophyta

- Lycophytes are terrestrial and some are epiphytes.
- The dominant plant is sporophyte.
- They produce upright stems and ground hugging stems.
- In upright stems small leaves can be found.
- Ground hugging stems produce dichotomously branching roots.
- They have strobili. In many club mosses and spike mosses sporophylls are clustered into club shaped cones/ strobili.
- They are homosporous or heterosporous.

Spike mosses are usually relatively smaller and often grow horizontally.

- All Club mosses are homosporous;
e.g. *Lycopodium* sp.
- All Spike mosses are heterosporous.
e.g. *Selaginella*



In some species the tiny gametophyte live above the ground and are photosynthetic. Others live below the ground and are nourished by symbiotic fungi.

Phylum Pterophyta

- Most sporophytes have rhizome (an underground stem)
- At the tip of the rhizome they produce leaves, called fronds
- Many fronds are highly dissected and feathery.
- All species are homosporous which develop into a bisexual gametophyte.
- Sporophytes are dominant
e.g. *Nephrolepis*.

Seed plants

Vascular plants consist of seeds and are called “seed plants”. They represent the vast majority of living plant species.

Seed plants can be divided into two groups, based on the absence or presence of enclosed chambers in which seeds mature.

1. Gymnosperms
2. Angiosperms

Gymnosperms are “naked seed” plants as their seeds are not enclosed in chambers.

Angiosperms are “seed plant” group accommodating all flowering plants. Angiosperm seeds develop inside chambers called ovaries; which originate within flowers and mature into fruits.

**Significant features of seed plants****1. Production of seed**

A seed consists of an embryo and endosperm. Endosperms supply food to the embryo. This endosperm is surrounded by a protective coat which is known as seed coat. When seeds are mature they can be dispersed through various dispersal methods.

Seeds are key adaptations that help seed plants to become the dominant producers on land and to exhibit the vast plant diversity of today.

2. Reduced gametophyte

The evolutionary trend of gametophyte reduction continued further in the vascular plants and led to seed plants. The gametophyte of seed vascular plants is not visible to the naked eye. They are mostly microscopic.

Tiny gametophytes develop from spores and are retained within the sporangia of sporophyte. This arrangement protects the gametophyte from environmental stresses. The moist reproductive tissues of the sporophyte shield the gametophyte from UV radiation and protect them from drying out. This relationship also enables the dependant gametophyte to obtain nutrients from the sporophyte.

3. Heterospory

Seed plants are heterosporous (produce both megaspores and microspores). Each megasporangium has a single functional megaspore and each microsporangium contains a large number of microspores.

4. Production of ovules and eggs

Seed plants are unique in retaining the megasporangium within the parent sporophyte. A layer of sporophyte tissue called integument envelops which protects the megasporangium.

The entire structure, containing megasporangium, megaspore and integuments is called an ovule. Inside each ovule female gametophyte develops from a megaspore and produces one or more eggs.

5. Production of pollen and sperms

A microspore develops into pollen grain that consists of a male gametophyte enclosed within the pollen wall. The wall of the pollen is tough as it is made up of the polymer sporopollenin, which protects the pollen grain during pollination. The transfer of pollen grain towards the ovule is called pollination. When a pollen germinates, it gives rise to a pollen tube that discharges sperm (male gametes) into the female gametophyte located within the ovule.

Inside a pollen grain, a sperm producing male gametophyte is present. The sperms of seed plants do not require motility as they are carried directly in to the eggs by pollen tubes. Some gymnosperms retain the ancient flagellated condition, however flagella have been lost in the sperm of most gymnosperms and all angiosperms.

Phylum Gnetophyta

e.g. *Gnetum*

- Only gymnosperms have vessels in xylem
- Leaves of gymnosperms look like those of the flowering plants. Their seeds, look like fruits of angiosperms.



Phylum Cycadophyta

- They have palm like leaves and large cones
 - They have flagellated sperms similar to those of seedless vascular plants
- e.g. *Cycas*



Phylum Coniferophytae.g: *Pinus*

- Large trees are included such as Cyperes and Red woods.
- In conifers two types of spores are produced by separate cones.

**Phylum Anthophyta - Angiosperms**

Presence of flowers -

- Stamens produce microspores and these microspores develop into pollen grains
- These pollen grains contain male gametophyte/ gametes
- Carpels produce megaspores and these megaspores produce female gametophytes/ embryo sac
- Seeds are enclosed within the carpels
- Production of fruits- seeds are protected by fruits which help in their dispersal. This is one of the unique features of phylum Anthophyta. A fruit typically consists of a fertilized ovary and sometimes include other persistent floral parts. After fertilization, the ovary wall thickens and develops into fruit. Ovules develop as seed of the fruit. The fruit protects dormant seeds and aid in their dispersal.

Diversity of Angiosperms

The flowering plants (angiosperms) are divided into two groups based mainly on the number of cotyledons in their embryo.

These two groups are;

1. Monocotyledons- species with one cotyledons
2. Dicotyledons- species with two cotyledons

Table 3 .2- Comparison of monocots and Dicots

Class – Monocotyledoneae	Class – Dicotyledoneae
The embryos have only one cotyledon	Embryos have two cotyledons
Fibrous root system	Tap root system
Parallel veins in leaves	Reticulate veins in leaves
Flower parts are trimerous	Flowers are pentamerous or tetramerous
Perianth present in flowers (No distinct calyx and corolla)	Distinct calyx and corolla present in flowers

Pollen grains are with one opening/ aperture	Pollen grains are with three openings/ apertures
Vascular bundles in the stem do not have cambia and are scattered e.g. grasses, coconut, rice	Vascular bundles in the stem have cambia and arranged in a ring e.g. Rose, shoe flower, cucurbits

The diversity of organisms within the kingdom Fungi

Kingdom Fungi

Characteristic features of Kingdom Fungi

- Eukaryotic
- Cell walls are made up of chitin a strong but flexible polysaccharide.
- They are absorptive and heterotrophs - many of them secrete extra cellular enzymes which aid in the breaking down of complex molecules into small molecules.
- Different species live as decomposers, parasites or mutualistics.
- Few are unicellular, others forming multicellular filaments called hyphae .
- Septa can be found in hyphae. (division of hyphae into cells by septa – cross walls).
 - Septum has a hole which enables the movement of mitochondria, ribosomes, nuclei etc.
 - Fungi lack septa are known as coenocytic fungi (with many nuclei)
- Fungal hyphae produce mycelium
- Some fungi produce haustoria (to penetrate and absorb or exchange nutrients between plants and the fungi)
- Multicellular fungi produce mycelia. (a network of branched hyphae adapted for absorption of nutrition)
- They show sexual and asexual reproduction.
- They produce spores.

Characteristic features of Phylum Chytridiomycota

e.g.: *Chytridium*

- Aquatic or terrestrial .
- Some are decomposers while others are parasitic.

- Multicellular or unicellular when multicellular it is coenocytic.
- They produce zoospores which are flagellated.
- Cell walls are made up of chitin.
- Some of them form colonies with hyphae while others exist as single spherical cell.

Characteristic features of Phylum Zygomycota

e.g. *Mucor*, *Rhizopus*

- Most of them are saprotrophs and some of them are parasites or commensals.
- Mycelium is coenocytic and aseptate. Septa found only where reproductive cells are formed.
- Asexual reproduction: Produce sporangia in which genetically identical haploid spores are produced. Also by endospores produced in sporangia.
- Sexual reproduction: A Zygosporangium is produced which is a sturdy structure produced by plasmogamy and karyogamy. Zygosporangium is resistant to unfavorable environmental conditions.
 - o Zygosporangium is a multinucleated structure which is resistant to drying and freezing.
 - o They are metabolically inactive in adverse environmental conditions.
- Zygosporangium produces genetically diverse haploid spores when environmental conditions are favourable.

Characteristic features of Phylum Ascomycota

e.g. *Aspergillus*, *Saccharomyces*, *Penicillium*

- Marine or freshwater or terrestrial
- Parasitic or symbiotic.
- Most of them are decomposers.
- Unicellular or filamentous, multicellular.
- In asexual reproduction conidia are produced at the tip of the conidiophores which are specialized hyphae. (Exospores in clusters or chains)
- In sexual reproduction fusion of sexually differentiated gametangia takes place and produce sac like structure called asci.
- Ascospores are produced within asci. Generally there are eight ascospores are produced in each ascus.
- Most of these fungi produce ascocarps enclosing asci.

Characteristic features of Phylum Basidiomycota

e.g. *Agaricus*, Puffballs, Shell fungi

- They are Terrestrial.
- They are major decomposers and some are symbionts.
- Filamentous with septae and dikaryotic.
- Mycelium is the dominant stage of the life cycle.
- They produce fruiting bodies called basidiocarps during sexual reproduction. Produce basidia on the gills of the basidiocarp.
- Produce basidiospores on basidium and exogenous.

The diversity of organisms within the kingdom Animalia**Kingdom Animalia****Characteristic features of Kingdom Animalia**

- Multicellular
- Heterotrophic eukaryotes- they ingest food and digest them in the body using enzymes
- Cells of the animals are organized into tissues.
- Most of them reproduce by sexually.
- Some show radial symmetry and some others show bilateral symmetry.

Phylum Cnidaria

Characteristic features of each example are not necessary

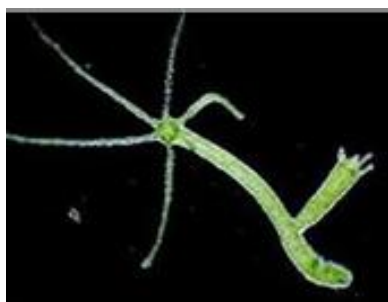
e.g. *Hydra*, Sea anemone, *Obelia*, Corals and Jelly fish



Obelia,



Jelly fish

*Hydra*

Coral polyp

- Majority of them are marine, except a few fresh water species. Some are macroscopic.
- Simple organization: diploblastic or just 2 body cell layers- an outer layers of ectoderm and inner layer of endoderm sandwiched between these two layers is a acellular layer of mesoglea
- They have a simple gastrovascular cavity which is a sac with a central digestive compartment. This cavity is lined by endodermis with a single opening (mouth) only.
- They show radial symmetry with two body forms polyp and medusa. Polyps are cylindrical forms attached to the substrate by the aboral end of the body. Tentacles are found around the mouth.
- Medusa resembles a flattened mouth down version of the polyps and they are free living.
- Some cnidarians exist only as polyps or only as medusa. Others have both polyp and medusa forms in their lifecycles.
- Tentacles are armed with cnidocytes which functioning defense and capturing prey.
- They have nematocysts which contain stinging thread.

Phylum Platyhelminthes

Characteristic features of each examples are not necessary

e.g. *Planaria*, *Taenia*, *Fasciola*

*Planaria**Fasciola*

- Commonly known as flatworms.
- Free living (*Planaria*) or parasitic (flukes and tapeworms).
- They are found in marine, fresh water and in damp terrestrial habitats.
- Body is dorsoventrally flattened. Some have elongated tape like body forms without true segmentation.
- Triploblastic with all three germ layers (ectoderm, mesoderm and endoderm). Signs of cephalization present but are not distinct.
- No body cavities, circulatory, respiratory and skeletal systems. The gaseous exchange is by simple diffusion through body wall.
- Sensory organs are found only in free living examples. Eye spots are found in the head.
- First appearance of little complex nervous and sensory system. A pair of anterior ganglion and two longitudinal nerve cords on central nervous system.
- Appearance of separate organs for excretion: Nitrogenous excretory system consists of protonephridia. These are a network of tubules with ciliated structures called flame bulb. These are used to maintain the osmotic balance.
- They have incomplete digestive system only with mouth without anus. Branched gastrovascular cavity is present for digestion. Some are having eversible pharynx.
- Free living examples have cilia for the locomotion.
- Some show asexual reproduction by regeneration. All are bisexual. Except tape worms (*Taenia*). Others have cross fertilization which is internal. In parasitic forms there are several larval stages, direct development in free living examples without larval stages.

*Taenia*

Phylum Nematoda (characteristic features of each example are not necessary)

e.g. round worms, hook worms, pin worms

*round worm**hook worm**pin worm*

- Most of them are free living in marine, few are fresh water and damp soil environments and parasitic in plants and animals.
- They are bilateral symmetrical. Triploblastic with pseudocoelomic .Their body forms are cylindrical with tapered ends. Body size varies from microscopic to macroscopic. They do not show distinct cephalization and segmentation. The sensory papillae are found on the anterior end of the body. Body is covered by tough cuticle and undergoes ecdysis.
- No circulatory and respiratory systems. Gaseous exchange is by simple diffusion through body wall. They have an Alimentary canal.
- Body wall is composed only of longitudinal muscles. They do not have special locomotary structures. Longitudinal muscles in the body wall are involved in locomotion.
- The Sexual reproduction is by internal fertilization. Sexes are separated and females are larger than males.

Phylum Annelida

Characteristic features of each examples are not necessary
e.g. Earthworms, Leeches and regworms.



Earth worm



Leech

- They can be marine, freshwater or in damp soil.
- They are segmented worms with cylindrical bodies
- They are Triploblastic.
- Coelom (true body cavity) is present for the first time.
- The first animals to show cephalization.
- Nervous system well developed; dorsal cerebral ganglion, ventral nerve chord, circumenteric connectives.
- Clitellum, Parapodia, setae and suckers are found in some examples. Clitellum is for external fertilization. Parapodia is used for locomotion and respiration. Setae are present for locomotion and suckers for locomotion and ingestion in ecto parasitic forms.

Phylum Mollusca

Characteristic features of each examples are not necessary

e.g. Oysters, Clams, Slugs, Snails, Octopus, Squids, Chitons and tusks shells



Squid



Octopus



Snail



clam



chiton



tusk shell



Oyster